

AP CALCULUS AB - UNIT 1

Guided Notes 1.8: Determining Limits Using the Squeeze Theorem



Name: _____ Class: _____ Date: _____

Learning Objectives and Essential Question

Essential question: How can we state all hypotheses of the Squeeze Theorem?

- State all hypotheses of the Squeeze Theorem.
- Create useful upper and lower bounds from absolute values and trigonometric inequalities.
- Distinguish persistent oscillation from shrinking oscillation.
- Write a complete AP-style squeeze justification.

Key Knowledge, Vocabulary, and Notation

Ordering Find functions $g(x) \leq f(x) \leq h(x)$ on a deleted neighborhood of the target.

Shared outer limit If $\lim g = \lim h = L$, then $\lim f = L$.

Absolute-value route If $|f(x) - L| \leq B(x)$ and $B(x) \rightarrow 0$, then $f(x) \rightarrow L$.

Bounded times vanishing If $|u(x)| \leq M$ and $v(x) \rightarrow 0$, then $u(x)v(x) \rightarrow 0$.

One-sided care Inequalities must hold on the side being considered; multiplying by negative quantities reverses order.

Concept Development and Strategy

1. Start from a known bound such as $-1 \leq \sin u \leq 1$ or $|\cos u| \leq 1$.
2. Multiply by a nonnegative shrinking factor or move to an absolute-value inequality.
3. End by naming the theorem and the common outer limit; merely saying “squeezed” is incomplete.

Worked Example: Shrinking oscillation

Evaluate $\lim_{x \rightarrow 0} x^2 \sin(1/x)$.**Answer and Reasoning**Because $|\sin(1/x)| \leq 1$, $|x^2 \sin(1/x)| \leq x^2 \rightarrow 0$. Therefore the limit is 0 by the Squeeze Theorem.

Worked Example: Shifted center

Evaluate $\lim_{x \rightarrow 3} (x - 3) \cos\left(\frac{1}{x-3}\right)$.**Answer and Reasoning**The absolute value is at most $|x - 3|$, which tends to 0. Thus the limit is 0.

Worked Example: Sine anchor from geometry

Given $\cos x \leq \frac{\sin x}{x} \leq 1$ for small positive x , find the right-hand limit at 0.

Answer and Reasoning

Both outer functions approach 1, so $\lim_{x \rightarrow 0^+} \sin x/x = 1$. Symmetry extends this to the two-sided limit.

Guided Practice and Discussion

Directions

Show the mathematical setup, preserve exact values unless a decimal is requested, and justify existence or interpretation statements with the appropriate definition or theorem.

1. Given $-x^2 \leq f(x) \leq x^2$ near 0, find $\lim_{x \rightarrow 0} f(x)$.

2. Evaluate $\lim_{x \rightarrow 0} x \sin(1/x)$.

3. Evaluate $\lim_{x \rightarrow 0} |x|^{3/2} \sin(7/x)$.

4. If $|f(x) - 4| \leq 3|x - 1|$ near 1, find $\lim_{x \rightarrow 1} f(x)$.

5. Evaluate $\lim_{x \rightarrow 0} \frac{x^3}{|x|} \sin(1/x)$.

6. Suppose $|f(x)| \leq |x|$ and $|g(x)| \leq x^2$ near 0. Find $\lim_{x \rightarrow 0} \frac{f(x)g(x)}{x}$ if defined for $x \neq 0$.

7. Prove: if u is bounded near c and $v(x) \rightarrow 0$, then $u(x)v(x) \rightarrow 0$.

AP Reasoning Check

Multiple-Choice Check

Choose the best answer. Explain why at least one distractor is incorrect during class discussion.

1. If $-2|x - 1| \leq f(x) \leq 5|x - 1|$ near 1, then $\lim_{x \rightarrow 1} f(x)$ is

- (A) -2 (B) 0
(C) 5 (D) not determined

2. Which bound is useful for $x^3 \cos(1/x)$ near 0?

- (A) $|x^3 \cos(1/x)| \leq |x|^3$ (B) $|x^3 \cos(1/x)| \geq 1$
(C) $\cos(1/x) \rightarrow 0$ (D) $x^3 \geq 1$

3. A graphing calculator shows violent oscillation for $x \sin(1/x)$ near 0, but the envelope narrows to the axis. The limit is

- (A) -1 (B) 0
(C) 1 (D) DNE

Common Errors and AP Precision

- Keep the value of a function at a point separate from its limiting behavior near that point.
- State one-sided behavior explicitly whenever a two-sided limit or continuity claim depends on agreement from both sides.
- A complete AP justification names the relevant definition, condition, or theorem rather than reporting only a numerical result.

Exit Ticket

Find $\lim_{x \rightarrow 0} x^2 \sin(1/x) \cos(e^{1/x^4})$ and justify every bound.

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